

An Improved Microcontroller-Based Sensorless Brushless DC (BLDC) Motor Drive for Automotive Applications

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Abstract—The direct back-electromotive-force (EMF) detection method previously described in a sensorless brushless dc (BLDC) motor-drive system (*Proc. IEEE APEC*, 2002, pp. 33–38) synchronously samples the motor back EMF during the pulsewidth-modulation (PWM) off time without the need to sense or reconstruct the motor neutral. Since this direct back-EMF-sensing scheme requires a minimum PWM off time to sample the back-EMF signal, the duty cycle is limited to something less than 100%. In this paper, an improved direct back-EMF detection scheme that samples the motor back EMF synchronously during either the PWM on time or the PWM off time is proposed to overcome the problem. In this paper, some techniques for automotive applications, such as motor-rotation detection, and current sensing are proposed as well. Experimental results are presented.

Index Terms—Back-electromotive-force (EMF) detection, current sensing, motor-rotation detection, sensorless brushless dc (BLDC).

I. INTRODUCTION

THERE ARE many motors inside a modern car with the majority of them brushed dc motors. In recent years, the brushless dc (BLDC) motor is receiving more interest for automotive applications. This is due to the higher reliability/longevity, lower maintenance, and quieter operation that BLDC has compared to its brushed dc counterpart. Over the last decade, continuing improvements in power semiconductors and controller ICs as well as the permanent-magnet brushless motor production have made it possible to manufacture reliable cost-effective solutions for a broad range of adjustable speed applications. As matter of fact, BLDC motors are being designed in or currently used in power steering, engine cooling fan, fuel/water pump, air-conditioning compressor, and heating, ventilating, and air-conditioning (HVAC) blower motors.

For three-phase BLDC motors, six-step commutation with 120° conduction time allows the current to flow in only two phases at any one time. This leaves the third phase available for sensing back EMF, which indicates the rotor position. Since the back EMF is directly related to the rotor position, sensing the back EMF will enable the controller to drive the motor

without Hall or other type position sensors. The back-EMF zero crossing of this third floating phase can be used to determine the commutation sequence. Both Erdman [2] and Uzukata and Uzuhashi [3] originally proposed the method of sensing back EMF to build a virtual neutral point that will, in theory, be at the same potential as the center of a Y wound motor and then to sense the difference between the virtual neutral and the voltage at the floating terminal. However, when using a chopping drive, the pulsewidth-modulation (PWM) signal is superimposed on the neutral voltage, inducing a large amount of electrical noise on the sensed signal. Moreover, the magnitude of the signal is often outside the common mode input range of the comparator used to sense the back EMF. To sense this properly requires a lot of filtering and attenuation. The filtering causes unwanted delay in the signal. The attenuation is required to bring the signal down to the allowable common mode range of the sensing circuit. The result is a poor signal-to-noise ratio of a very small signal, especially at startup where it is needed most. Consequently, this method tends to have a narrow speed range and poor startup characteristics. To reduce the switching noise, the back-EMF integration [4] and third harmonic voltage integration [5] were introduced. The integration approach has the advantage of reduced switching-noise sensitivity. However, they still have a problem at low speed since attenuation cannot be avoided.

A direct back-EMF-sensing scheme in which back-EMF signals can be directly extracted for each phase without sensing the motor neutral point is presented in [6]. This is done by synchronously detecting the back EMF of the floating phase during the high-side-switch PWM off time. This back-EMF-sensing scheme has advantages over the conventional method [2]–[5]. There is no attenuation in the back-EMF signal, and no high common mode voltage issue since the motor neutral voltage is not used for zero-crossing comparison. Precise back-EMF zero-crossing detection can be achieved without low pass filtering. As a result, this control method has wider speed range and better startup performance.

This direct back-EMF-sensing scheme has been implemented into a low-cost microcontroller [6], and it has been used for various applications [7], [8]. Certain improvements of this method, such as reducing power loss in the MOSFETs for low-voltage applications, reducing time delaying in high-voltage applications, were reported in [8]. However, since this direct back-EMF-sensing scheme requires a minimum PWM “off” time, the duty cycle is limited to something less than 100%. In many automotive applications, the desire is to run the

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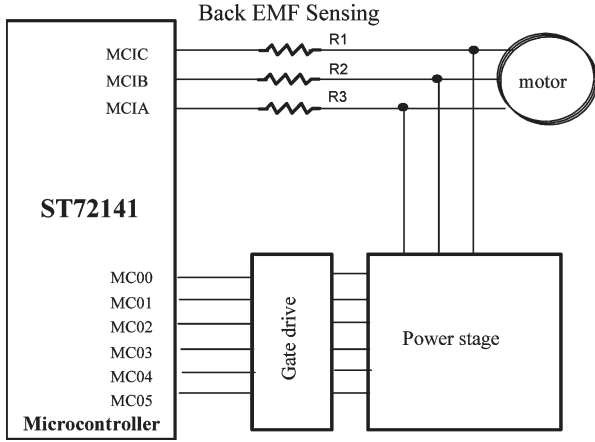


Fig. 1. Direct back-EMF-sensing block diagram.

motor at 100% duty cycle to fully utilize the low bus voltage. An improved direct back-EMF detection scheme that samples the motor back EMF synchronously during either the PWM off time or the PWM on time is proposed in the paper to overcome this problem.

In automobiles, windmilling effect can make fans rotate without electric power. When the controller needs to control the motor, if the motor is already spinning, the controller should be able to determine if the motor is rotating and in what direction. In this paper, a method for the microcontroller to detect the motor rotation is presented. Also, a current-sensing method for protection without a current-sensing resistor is proposed in this paper as well.

II. REVIEW OF DIRECT BACK-EMF SENSING FOR BLDC DRIVES [6]

Generally, a BLDC motor is driven by a three-phase inverter with what is called six-step commutation. In the reference scheme [6], the PWM signal is applied to the high side switches only. The back-EMF signal can be synchronously sampled on the unused phase during the high-side-switch PWM off time. The back-EMF signals are directly fed into the microcontroller through resistors, as shown in Fig. 1. In this scheme, only three resistors that limit the injected current are needed for back-EMF sensing.

Assuming at a particular step, phases *A* and *B* are conducting current, and phase *C* is floating. The upper switch of phase *A* is pulsewidth modulated to regulate motor speed or current. The lower switch of phase *B* is on during the entire step, which is switched only at commutation. The terminal voltage ν_c is sensed when the upper switch is off. When the upper switch of phase *A* is turned on, the current is flowing through the switch to windings *A* and *B*. When the upper transistor of the half bridge is turned off, the current freewheels through the diode paralleled with the bottom switch of phase *A*, as shown in Fig. 2. During this freewheeling period, the terminal voltage ν_c is detected as phase *C* back EMF when there is no current in phase *C*.

From the circuit in Fig. 2, it is easy to see that $\nu_c = e_c + \nu_n$, where ν_c is the terminal voltage of the floating phase *C*, e_c is the phase back EMF, and ν_n is the neutral voltage of the motor.

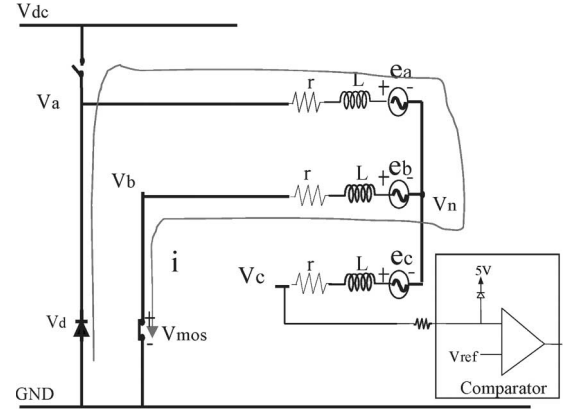


Fig. 2. Back-EMF detection during the PWM off-time moment.

From phase *A*, if the forward voltage drop of the diode is ignored, we have

$$\nu_n = 0 - ri - L \frac{di}{dt} - e_a. \quad (1)$$

From phase *B*, if the voltage drop on the switch is ignored, we have

$$\nu_n = ri + L \frac{di}{dt} - e_b. \quad (2)$$

Adding (1) and (2), we get

$$\nu_n = -\frac{e_a + e_b}{2}. \quad (3)$$

Assuming a balanced three-phase system, if only the fundamental frequency is considered, we have

$$e_a + e_b + e_c = 0. \quad (4)$$

From (3) and (4)

$$\nu_n = \frac{e_c}{2}. \quad (5)$$

So, the terminal voltage ν_c

$$\nu_c = e_c + \nu_n = \frac{3}{2}e_c. \quad (6)$$

From the above equations, it can be seen that during the PWM off time, which is the current freewheeling period, the terminal voltage of the floating phase is proportional to the back-EMF voltage without any superimposed switching noise. It is also very important that this terminal voltage is directly referred to the ground instead of the neutral point. So, the neutral-point voltage information is not needed to detect the back-EMF zero crossing. Since the true back EMF is extracted from the motor terminal voltage without any filtering, the zero crossing of the phase back EMF can be detected very precisely. Synchronously sampling the back-EMF signal near the end of the PWM off time allows any switching noise to settle out and allows sampling the signal without any filtering and the associated delay.

However, it requires a minimum PWM off time of a few microseconds to detect the back EMF. This limits the maximum

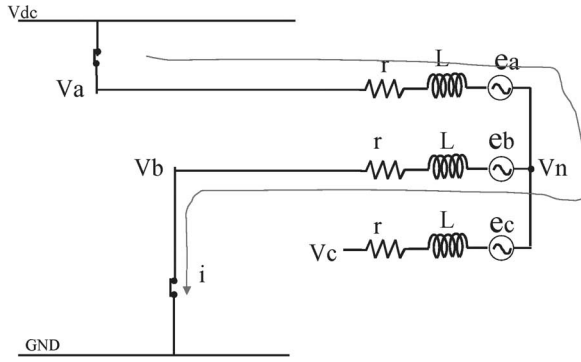


Fig. 3. Winding terminal voltage during the PWM on time.

duty cycle to around 90% for 20-kHz switching frequency. In some automotive applications, it is desired to be able to run the motor at 100% duty cycle to fully utilize the low battery voltage. Then, the voltage constant of the motor can be increased by 10%. It will benefit the motor design and improve the overall system efficiency.

If the back-EMF detection can be done during PWM on time, then the limitation on the maximum duty cycle can be removed, and the motor can be driven with the full 100% duty cycle. An improved back-EMF detection is described in the following section.

III. IMPROVED DIRECT BACK-EMF-SENSING SCHEME: DETECT THE BACK EMF DURING THE PWM ON TIME

We can derive the floating-phase terminal winding voltage during the PWM on time. As shown in Fig. 3, it is assumed that phases A and B are conducting current; phase C is floating.

From phase A, we can derive the value of ν_n

$$\nu_n = \nu_{dc} - ri - L \frac{di}{dt} - e_a. \quad (7)$$

From phase B, we can derive the value of ν_n

$$\nu_n = ri + L \frac{di}{dt} - e_b. \quad (8)$$

From (7) and (8), we derive

$$\nu_n = \frac{\nu_{dc}}{2} - \frac{e_a + e_b}{2}. \quad (9)$$

In a balanced three-phase system, if only the fundamental frequency is considered, we have

$$e_a + e_b + e_c = 0. \quad (10)$$

Incorporating (10) into (9), we obtain

$$\nu_n = \frac{\nu_{dc}}{2} + \frac{e_c}{2}. \quad (11)$$

So, the terminal voltage ν_c can be expressed by

$$\nu_c = e_c + \nu_n = \frac{3}{2}e_c + \frac{\nu_{dc}}{2}. \quad (12)$$

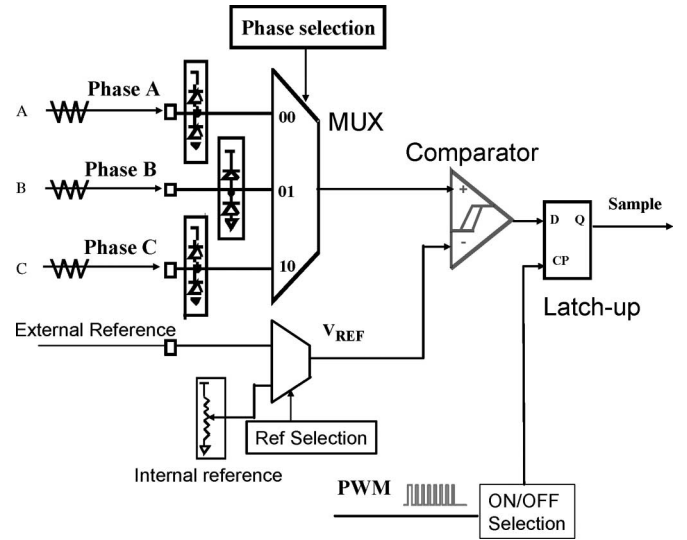


Fig. 4. Hardware implementation for improved back-EMF detection.

Therefore, during the high-side-switch PWM on time, the back-EMF zero-crossing information can be obtained by comparing the phase voltage with $\nu_{dc}/2$. By sensing the back EMF during high-side-switch PWM on time, it is possible to extend the duty cycle to 100%. In many cases, it will require attenuation to bring the $\nu_{dc}/2$ to be within the common mode voltage range of the comparator. This should not be a problem since the amplitude of the back-EMF signal is high during high-speed operation. It is still preferred to use back-EMF detection during PWM off time for startup because there is no signal attenuation. Therefore, we can combine two sensing schemes together to optimize the system. That is, during startup and low-speed (e.g., < 50% duty cycle) operation, the back EMF can be detected during PWM off time, and then during high-speed (e.g., > 50% duty cycle) operation, the back EMF can be detected during PWM on time.

IV. IMPLEMENTATION AND EXPERIMENTAL RESULTS

Fig. 4 shows the hardware implementation for this combined direct back-EMF detection scheme.

According to the commutation sequence, the controller selects which phase back EMF to sense via the multiplexer. The multiplexer passes the selected phase back-EMF signal to the sense comparator. The selected phase voltage is compared to a reference for determining back-EMF zero crossing and is latched accordingly. This reference can be either internal or external and is selectable according to which back-EMF-sensing scheme is used. The comparator output can be latched during either PWM on time or off time again, which is selectable by software. When the back-EMF zero crossing is detected during the PWM off time, the internal fixed reference, which is close to 0 V, is selected. When the back-EMF zero crossing is detected during the PWM on time, the external reference is selected. The external reference voltage would be proportional to $\nu_{dc}/2$. The software can make the transition smoothly by looking at duty cycle.

The improved direct back-EMF-sensing hardware integrated with a standard low-cost specific application microcontroller,

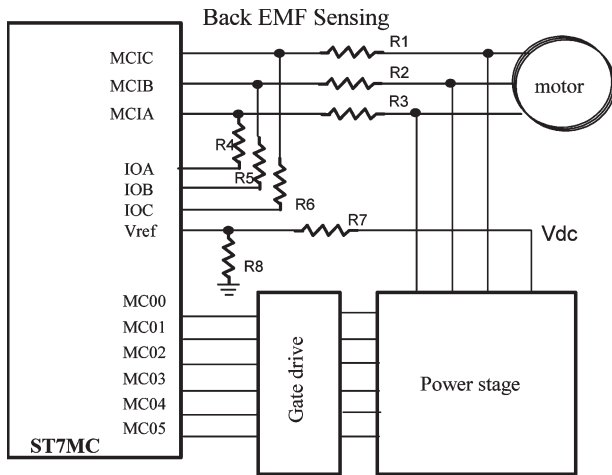


Fig. 5. Implementation of improved direct back-EMF-sensing scheme.

which is the ST7MC, is developed. Fig. 5 shows a system block diagram of the microcontroller system that can do both back-EMF detection schemes during the high-side-switch PWM on or off time.

During startup and at low PWM duty cycles, the controller is configured to detect back EMF during the PWM off time. During PWM off time, the back-EMF zero-crossing event voltage is at or near 0 V, so the reference voltage is set to the fixed low internal voltage. With that, the input/output A (IOA), input/output B (IOB), and input/output C (IOC) pins are configured as high-impedance inputs to prevent the attenuation generated by the R1/R4, R2/R5, and R3/R6 resistor pairs. This allows the controller to see the actual back-EMF voltage without attenuation thus improving low-speed sensitivity and startup performance.

At higher duty cycles, the controller is configured to detect back EMF during the high-side-switch on time. Here, attenuation is needed to bring the voltage within the common mode range of the comparator as the back-EMF zero-crossing event is centered on $v_{dc}/2$. With that, the IOA, IOB, and IOC pins are reconfigured as outputs and set to logic low. All back-EMF signals are then attenuated by the R1/R4, R2/R5, and R3/R6 resistor pairs. The microcontroller will select the external reference, the attenuated dc bus voltage, for back-EMF zero-crossing detection.

Figs. 6 and 7 show waveforms for the improved detection scheme that the back EMF is sampled during PWM off time and on time.

In Fig. 6, the back EMF is detected during PWM off time; there is no attenuation of the signal, but the peak of the signal is clamped at 5 and -0.7 V by the internal protection diodes. Since only the back EMF of the floating winding is detected, the sensing pin of the microcontroller is tied to ground when the winding is conducting to reduce the noise. This is shown in Fig. 6; during the period that the winding is energized with PWM, the sensing signal is low. For the zero-crossing signal, which is shown on the second trace, every toggling transition corresponds to one zero crossing detected.

In Fig. 7, the back EMF is detected during PWM on time, the signal is attenuated. Since the amplitude of the signal is high

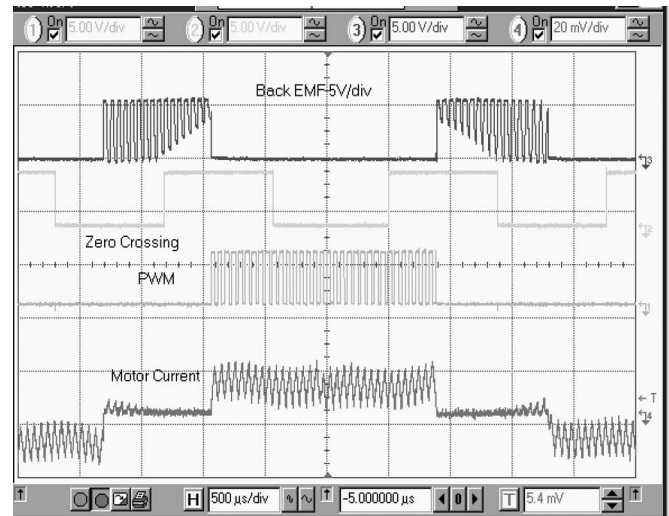


Fig. 6. Key waveforms for back-EMF sensing during PWM off time.

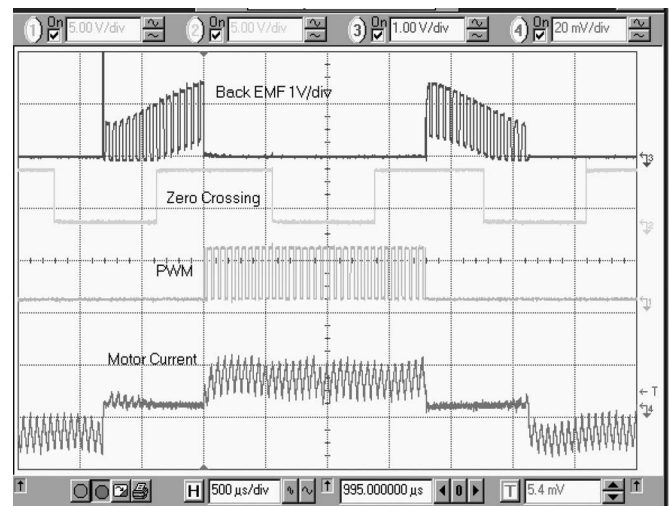


Fig. 7. Key waveforms for back-EMF sensing during PWM on time.

during high-speed operation, the attenuation is not a problem. Please note the different back-EMF shape in Figs. 6 and 7.

Fig. 8 demonstrates a running system at 100% duty cycle. The waveforms show the terminal voltage, current, and zero-crossing signal for same phase. From this waveform, it can be seen that the back-EMF zero crossing happens at half of the dc-bus voltage.

A. Startup

The sensorless schemes are not self-starting. In order to sense the back EMF, the motor must first be started and brought up to a certain speed where the back-EMF voltage can be detected. In practice, open-loop starting [10] is accomplished by providing a rotating stator field with a certain increasing frequency profile. After the speed reaches certain level, the back EMF can be detected, providing the position information and the system switches to synchronous commutation mode.

If there is no specific requirement for startup or light load, like fan applications, this open-loop startup can be satisfactory. However, for some applications, e.g., automotive fuel pumps,

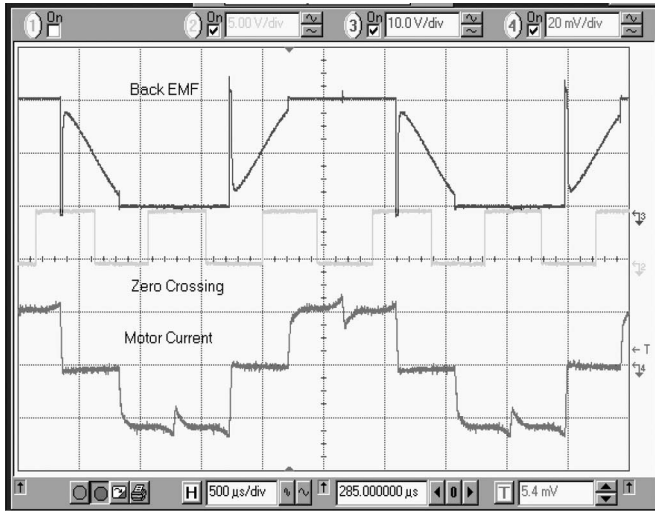


Fig. 8. Running system at 100% duty cycle.

the startup has to be very fast to ensure that there is sufficient fuel pressure within a very short time. It is very difficult to tune the startup using the open-loop starting algorithm. In [7] and [11], an optimized startup routine by manually tuning the commutation time with the help of a tachometer is described. According to the feedback of the tachometer, each commutation step time during startup is set such that the motor speed reaches maximum value at the end of that step. This method is also suitable for medium-to-heavy-load startup condition.

V. MOTOR-ROTATION DETECTION AND CURRENT SENSING

A. Motor-Rotation Detection

Sometimes, the motor is rotating before the controller starts the motor. For example, in an automotive engine cooling fan, there can be a windmilling effect that could cause the rotor to rotate without electrical power. A rotating rotor generates detectable back EMF. Once back EMF is detected, rotation can be determined. If the rotor is rotating in the right direction, there is no need to perform the startup. If the rotor is rotating in the wrong direction, then the microcontroller should be able to sense this and make the proper corrections to start the motor properly in the right direction.

If the system is configured, as shown in Fig. 1, the back-EMF signals are shown in Fig. 9 when the motor is turning because of the windmilling effect.

In an electrically undriven motor, where the rotor is rotating by some external means, the back-EMF signals are floating. With no reference to the motor neutral point, the microcontroller is prevented from detecting the back-EMF zero crossing. To provide a reference point for the microcontroller, three other resistors (R_n) are added, as shown in Fig. 10. As a result, the back-EMF signals will change to those shown in Fig. 11.

In Fig. 11, the zero crossings of the three-phase back-EMF signals are clearly identifiable. At rising edge of v_a , according to the sign of v_b and v_c , the rotation direction can be determined. For example, if the motor is turning in right direction, the sequence of three phases is $A-B-C$; otherwise, if the motor

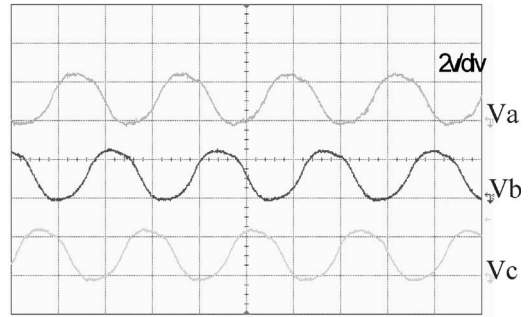


Fig. 9. Back-EMF signals when motor is rotating by windmilling effect.

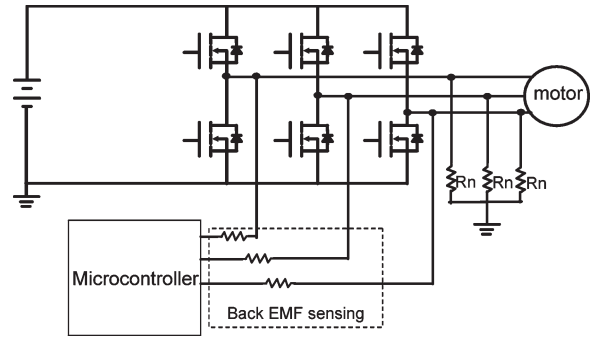


Fig. 10. Three resistors R_n are added in the winding terminals.

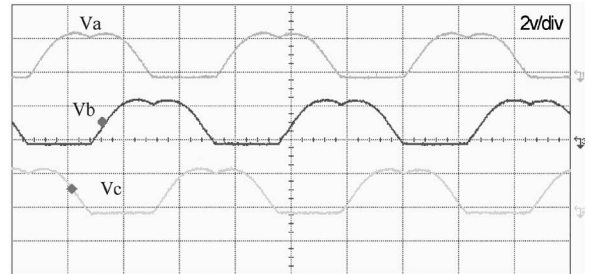


Fig. 11. Back-EMF signals after adding three terminal resistors.

is turning backward, the sequence will be $A-C-B$. Since the motor is turning in right direction, as shown in Fig. 11, at rising edge of v_a , v_b is low and v_c is high. If there is no signal at v_a , v_b , and v_c , the motor is not running.

If the micro detects that the motor is turning in right direction, zero crossing of the back EMFs is available for the rotor position. Therefore, the controller can drive the motor directly in synchronous mode according to the zero-crossing information without regular startup routine. The controller can estimate the motor speed and give PWM duty cycle accordingly. If the motor is turning in wrong direction, the controller has to brake the motor by shorting three bottom MOSFET or injecting dc braking current. After the motor stops, the controller can perform the regular startup routine.

B. Current Sensing

To protect the circuit from overcurrent conditions or rotor stall condition, some form of current sensing is required. The most common solution is to place a low-ohm current-sensing resistor in the current path at one of the power rails (ground

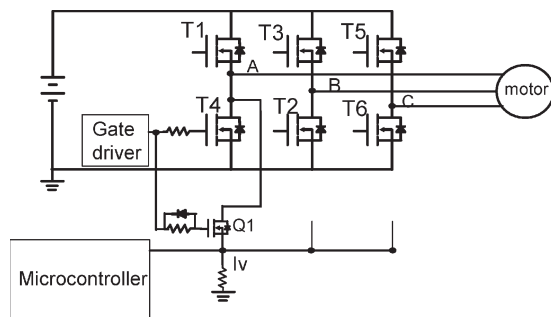


Fig. 12. Current-sensing circuit.

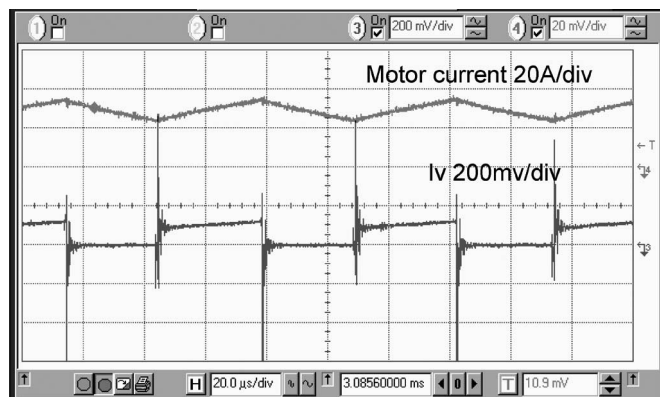


Fig. 13. Motor current and voltage signal from MOSFET.

or supply) [7]. The current-sensing resistor not only causes additional power losses, but also it brings some difficulty in circuit layout and packaging.

A solution can be realized by incorporating the properties of the MOSFET switches typically used in the inverter. MOSFETs have a linear resistance characteristic when driven fully on. It is possible to get the current information from the voltage drop seen across a fully enhanced MOSFET switch. This current information is only used for current-protection purposes, therefore, the variation of $R_{ds(on)}$ is acceptable and in some ways beneficial.

A simple circuit is proposed to perform the current detection by measuring the low side MOSFET voltage drop. Fig. 12 shows circuit for only one phase. The other two phases are exactly the same. When the MOSFET T4 is turned on, Q1 is turned on. The voltage of point Iv is the voltage drop across the MOSFET T4. When T4 is turned off, Q1 is turned off, too. The voltage of point Iv will be zero. Fig. 13 shows the waveforms.

VI. CONCLUSION

The original direct back-EMF-sensing scheme has a maximum duty-cycle limitation, since there is a required high-side-switch minimum PWM off time to do the detection. The improved direct back-EMF-sensing scheme eliminates this duty-cycle limitation by adding the option of sensing the back EMF during the high-side-switch PWM on time. An optimized system uses both sensing schemes where during motor startup and low speed, when back EMF is low, it is preferred to use the original scheme, since there is no signal attenuation. Then, at higher speeds, where the back EMF is much easier to detect,

the system can switch to the alternative back-EMF-sensing scheme introducing some attenuation to the back-EMF signal and allowing for the high side switch to be full on.

With the combination of two detection schemes in one system, the motor commutation detection can be optimized over the entire speed range. The experimental results verify this concept and show a good performance.

For automotive applications, the algorithm to detect motor rotation caused by the windmilling effect is very useful. Also, the method of measuring voltage drop on MOSFET can provide over-current protection for the circuit but without current-sensing resistor.

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